

CHAPTER 3 SITE REQUIREMENTS

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Operational environment

This section outlines the site requirements for a Mastersizer 3000. Make sure all of these are met before the Malvern Panalytical engineer arrives to install and commission the system.

Environmental conditions

The site must be:

- Away from strong light sources (e.g. windows).
- Away from strong heat sources (e.g. radiators).
- Well ventilated (for noxious samples).
- On a horizontal vibration-free bench built to support the weight of the system.

Table 3.1 Environmental conditions

Unit	Weight	Model number
Mastersizer 3000 / 3000E	30 kg	MAZ3000 / MAZ3010
Hydro SV	4.55 kg (3.05 kg + 1.5 kg)	MAZ3100
Hydro SM	9.75 kg (1 kg + 8.75 kg)	MAZ3150
Hydro MV	5 kg	MAZ3200 / MAZ3210
Hydro LV	5 kg	MAZ3300 / MAZ3310
Hydro EV	5 kg	MAZ3400
Hydro wet cell	2.46 kg	
Aero S / M	10.5 kg	MAZ3500 / MAZ3550
Aero dry cell	2 kg	

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Store/operate the system in the following conditions (accurate measurements are sample-dependent, for example dry powders may stick together in high humidity):

Table 3.2 Storage and operational conditions

IP rating	IP41B
Operational conditions	5°C to 40°C (41°F to 104°F)
Storage conditions	-20°C to 50°C (-4°F to 122°F)
Humidity	Maximum humidity 80% for temperatures up to 31°C, decreasing linearly to 50% relative humidity at 40°C
Usage	Indoor use only
Altitude	Up to 2000 m
Mains supply voltage fluctuations	Up to $\pm 10\%$ of nominal voltage
Overvoltage category	II (IEC 60664)
Pollution degree	2 (BS EN 60664-1:2003)
Installation category	II (BS EN 60664-1:2003)

In addition:

- Do not obstruct power sockets as they may need to be disconnected during an emergency.
- Avoid passing electrical cables through areas where liquids can be spilled.
- Pipes should always be routed through the pipe porting on the bottom of the instrument as shown in the illustration:

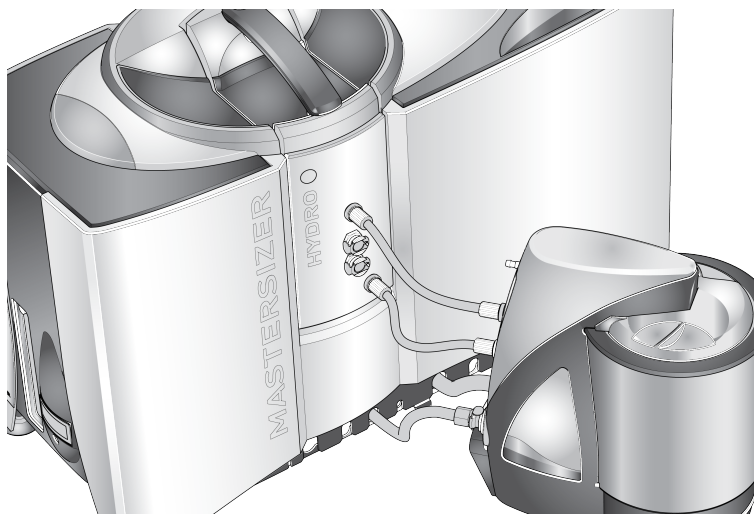


Figure 3.1 Instrument tubing.

Space required for system

Provide enough space to allow easy access to all components of the system. Allow at least 800 mm above the bench surface for access to the cell and dispersion unit. Any drain pipes from the Hydro LV/MV or SM units must not exceed **2.0 m** in length.

Table 3.3 Space required

Component	Length	Width	Height
Optical bench	690 mm	300 mm	450 mm
Hydro SV	215 mm	110 mm	275 mm
Hydro SM			
— Dispersion unit	140 mm	175 mm	390 mm
— Dispersion unit controller	225 mm	170 mm	70 mm
Hydro LV / MV	280 mm	180 mm	300 mm
Hydro EV	220 mm	150 mm	300 mm
Aero S / M	260 mm	180 mm	380 mm

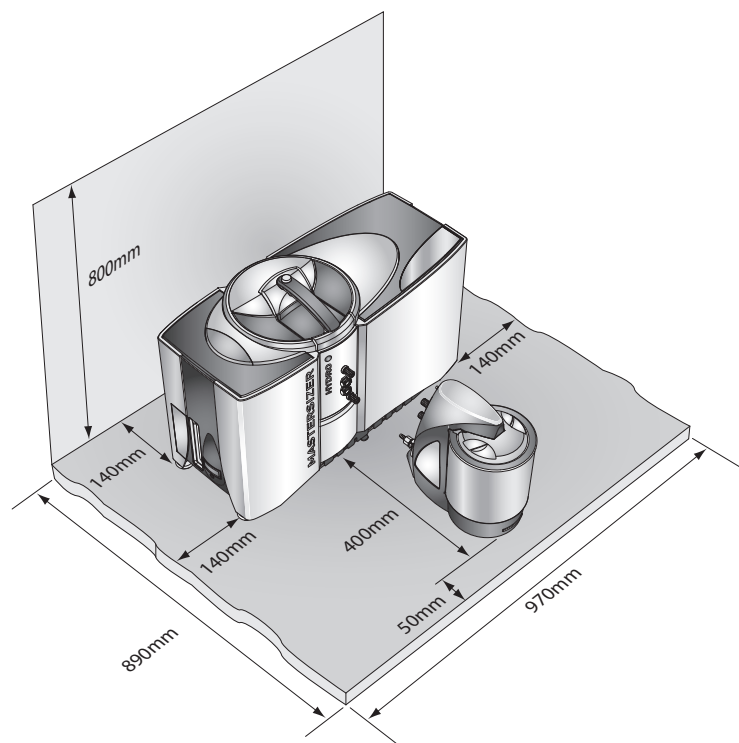


Figure 3.2 Minimum recommended space for a wet system.

Services required

This section describes the services that are required for installing the Mastersizer 3000 system.

General

Table 3.4 General requirements

Requirement	Optical bench	Hydro SV	Hydro SM	Hydro MV	Hydro LV	Hydro EV	Aero S/M
Power sockets	1	-	-	-	-	-	1 (a)
Direct plumbed dispersant (option)	No	No	No	Yes	Yes	No	No
Drain	No	No	Yes	Yes	Yes	Yes (b)	No
Vacuum unit	No	No	No	No	No	No	Yes
Air line	No	No	No	No	No	No	Yes
Efficient extraction system	No	No	No	No	No	No	Yes

In addition, the computer system requires one or more power sockets.

- The ancillary switching unit for the vacuum requires a power socket.
- The drain does not need to be within 2 m on the Hydro EV or SV. The EV uses a beaker and the SV a cuvette to hold the sample. Both are emptied by the operator.

Power specification

The mains power supply must be clean and filtered. If necessary, fit an uninterruptible power supply (UPS) to remove any spikes or noise.



Note:

The instrument provides power directly to the connected dispersion unit via the CAN connection. The power requirement is given below:

Mastersizer 3000 optical unit

- 50 W — with no dispersion units connected.
- 200 W (maximum) — with 2 dispersion units connected.
- The Mastersizer 3000 can supply up to 96 W to any one dispersion unit, and up to 150 W in total to the dispersion units fitted.



Note:

Only use the power supply unit (PSU) and cables provided. Using another PSU voids any warranty and may be unsafe.

Services specification

Direct plumbed dispersant specification (Hydro LV/MV only)

For most applications, the liquid dispersant is water from a header tank or the mains supply. If the input pressure to the dispersion unit exceeds 4 bar g, an external pressure regulator **must be fitted** to reduce this to 4 bar g.

The dispersant input must be filtered to remove particulates. The filtering depends on the size of the sample; if the sample contains a significant proportion of sub-micron particles the dispersant input should be filtered to remove them.

Hydro MV

- Minimum input pressure: 1 bar g (100 kPa g) @ 25 mL/s-1
- Maximum input pressure: 4 bar g (400 kPa g) @ 25 mL/s-1

Hydro LV

- Minimum input pressure: 1 bar g (100 kPa g) @ 25 mL/s-1
- Maximum input pressure: 4 bar g (400 kPa g) @ 25 mL/s-1

Hydro wet cell water jacket requirements (Hydro LV/MV/EV/SM)

- The Maximum pressure for water jacket connections is 0.5 bar g.

Drain requirements

Always dispose of waste dispersant and sample mixtures responsibly. Many local laws forbid disposal of chemicals in ways that allow their entry into the water system. Seek local advice on disposal of chemical waste.

Hydro LV/MV/SM

To allow liquid to drain efficiently the drain/waste must:

- Be within 2 m of the dispersion unit
- Be lower than the bench surface
- Slope gently downwards
- Contain no loops or kinks

Discharge to a normal sink is sufficient, provided the sample and dispersant are non-hazardous.

It is important that the draining end of the drain pipe is above the level of the waste liquid at all times. It must not be under the surface of the liquid otherwise the dispersion unit will not drain efficiently.

Hydro SV/EV

The EV uses a beaker and the SV a cuvette to hold the sample. Empty either unit manually into the drain/waste, either directly or via a washstation (Hydro SV). This can be a normal sink, provided the sample and dispersant are non-hazardous.

Air line specification (Aero S/M)



WARNING!

The air pressure regulator fitted to the Aero has a maximum input pressure limit of 10 bar. If there is any possibility that the air supply could exceed this, fit a protection device.



CAUTION!

The air line supply must be dry, free from oil and filtered to less than 0.01 mm. Failure to meet the air line specification will permanently damage the dispersion unit and invalidate the warranty.

Air line specification

Table 3.5 Specification for the air line

Item	Specification
Air line pressure	5.5 to 8 bar g, but should nominally be set to 6 bar g
Air flow	At least 82 l/min at 6 bar (87 psi g).
Air line particulate filtering	To Class 1 of ISO8573.1:2010
Air line water filtering	To Class 4 of ISO8573.1:2010
Air line oil filtering	To Class 1 of ISO8573.1:2010

The outlet from the compressor must be fitted with a push-fit adapter capable of taking a 6mm o/d polyurethane pneumatic pipe.

Sound emissions from the system (including the compressor) must be less than 85 dBA, (measured at the operator's position). Malvern Panalytical can supply a compressor meeting this specification.

Vacuum unit specification (Aero S/M)

The exhaust from the Aero air cell must be extracted by an extraction system/vacuum unit.

**WARNING!**

Never let the exhaust discharge into an enclosed room.

If Malvern Panalytical does not provide the vacuum unit used with the Aero unit, any vacuum unit used must conform to the following specification (minimum specification before exhaust filter fitted).

Vacuum unit specification

Table 3.6 Specification for the vacuum unit

Parameter	Condition	Value
Power	Continuous	2.8 kW
Air flow	—	39 liters/sec
Vacuum	—	20.4kPa
Suction power	With hose fitted to air cell output port.	182 W
Diameter of Vacuum hose	38 mm (inside diameter)	To connect to vacuum port on MS3000 optical unit

Malvern Panalytical can supply a vacuum unit meeting this specification.

Note:

An adaptor cable is required to connect an IEC 60320 C13 power cable (as used by the Nilfisk GM80 vacuum cleaner) to the higher-rated IEC 60320 C20 connector on the ancillary switching unit.

This is available from your Malvern Panalytical representative.

Ancillary switching unit (Aero S/M)

The Ancillary switching unit is an external control accessory that is used to switch on the vacuum unit required for the sample extraction during each measurement. This must be used in conjunction with any vacuum unit.